



**Department of Electrical, Computer
& Biomedical Engineering**
Faculty of Engineering & Architectural Science

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Instructor:	Dr. Gul Khan
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Abstract:

The final course project for the SystemC-based Network-on-Chip design focuses heavily on simulating and modelling a fully operational Network-on-Chip(NoC) architecture which utilizes the extensive library of SystemC to efficiently execute communication between distinct IP cores within the System-on-Chip(SoC) environment. The design includes various components and subsystems working synchronously to execute the NoC tasks, specifically complex components like the routers that are equipped with arbiters, FIFO buffers, and crossbar switches which all work together to enable routing and coordination to effectively perform the packet-based communication using a wormhole flow control system. The project also involves the IP cores having a Source and a Sink; the Source sends packets(made up of number of flits) to the router to be distributed out to the network, and the Sink is the receiver for the IP core that appropriately receives the packets from the router in an orderly fashion. Now the objective of this project as of now is to understand the routing behavior, synchronization, and performance of the packet-based communication system, from aspects like packet delay. Multiple modifications were implemented to synchronize the clocks, measure transmission delays within the communication, and correctly fix the arbiter module's routing behavior to reflect a proper wormhole packet architecture where only the head flit leads routing as it has the desired destination information, and the body just follows through with the head flit.

Introduction:

It is understood that the Network-on-Chip(NoC) architectures replace the traditional bus-based communication

systems we see from old SoC chip designs, ultimately making modern SoCs much more scalable and efficient. This project very much simulates a small-scale fully functional version of a mesh NoC using SystemC, allowing analysis and documentation of packet transmission as well as the routing mechanism behaviours within the mesh NoC prototype. The design comprises modular components and subsystems, enabling the ability for the NoCs to all execute parallel communication between multiple processing elements, ultimately improving throughput and reducing congestion. This project focuses on designing and simulating a NoC using SystemC starting from a 1×2 mesh topology and extending it into a full operational 4×4 mesh(multiple NoCs working simultaneously).

Theoretical Explanation:

A NoC consists of an entire transmission or communication system that utilizes routing mechanisms to accurately coordinate packets to appropriate destinations within the network. The IP cores are equipped with routers that interconnect within the network, and communicate in the form of packets that are further divided into flits, that is the smallest transferable unit. It could be noted that the routing mechanism used in this project is a deterministic XY routing, where the arbiter component routes the direction based on the relative positions of the source and destination, using North, South, East and West as parameters.

Now each router contains a FIFO buffer for temporary storage of flits for further processing, then an arbiter unit that computes and coordinates the effective routing path typically focused on resolving contentions between same routing paths that could potentially lead to a deadlock & routing crashes. The router also consists of a crossbar switch for effectively executing routing between inputs and outputs. The entire design

is implemented within the wormhole flow control, where the header flit establishes the path by dictating the destination address, the subsequent body flits really just follow the same path without needing to reevaluate the routing decisions. The tail flit releases the reserved path, notifying that the packets have finished its route and destination is reached. This method is very effective as it reduces latency and buffer requirements compared to a redundant storing and forwarding single flits routing process.

The system design also implements backpressure and an organized FIFO control mechanism, solely using the acknowledgement signals for this, which confirms and ensures that the current data has reached the destination before continuing to transmit the rest of the data making sure the components are ready, ultimately preventing congestion and overflow.

Work Progress/Interim Questions:

1) The source module represents the sender/transmission unit responsible for generating packets and setting them with the appropriate data and packet values. These packets are then injected to the router to be distributed into the network. The source module contains input ports for its source_id, acknowledgement(ack_in), and clock along with its output port which is packet_out. The source_id is a unique identifier for the source as all the IP cores have their own source module. This module also operates utilizing a clock sequence that toggles the clock value each time the router reaches it, this ensures that packets are unique each time and not ignored due to it being the same; this makes sure the events are triggered appropriately. Do note that the source module is responsible for creating the incoming synthetic traffic.

It needs to be understood that the packets are

constructed at flit level. Each flit contains fields such as data, source ID, destination ID and a header/tail indicator. A single packet contains multiple flits(5 for this specific implementation right now), where the final flit is marked as the tail. The source module uses the input acknowledgement signal as an indicator that the data and flits have been received by the router, as the router sends back the acknowledgement when it is ready again to take in more flits; this implementation ensures that flow control is maintained and flits are not lost in transmission errors.

2) The theoretical router from figure 8 displays a 5x5 router structure with FIFOs, Demultiplexers units with each input port, and smaller intermediate stages that are not essential and necessary for the simulation-level design as the modular design accounts for such intermediate stages. The following router structure actually used for this project is much simplified; not all input ports are connected to the FIFO buffers, as they are not all needed to be connected, and as it could clearly be seen the demultiplexer structures are entirely omitted. Instead, it could be seen that the routing decisions and organization aspects are entirely handled by the arbiter and the crossbar switch module. These simplifications reduce the complexity while promoting a modular and clean programming where no extra submodules are required to complete the tasks; this puts emphasis on functionalities like buffering, arbitration and crossbar switching.

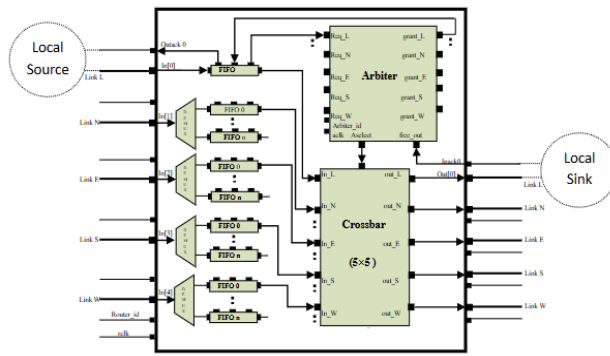


Figure 8: 5x5 Generic Router

Figure 1: Original 5x5 Router Layout

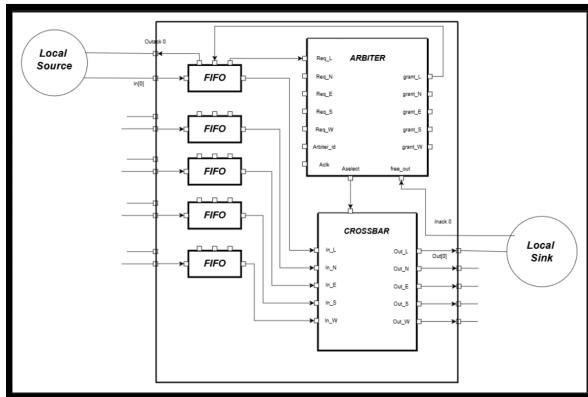


Figure 2: Modified Project 5x5 Router Layout

3) Now when the source and router clocks are synchronized, it so happens that the initial flits are not instantly received by the sink due to the improper initialization and synchronization of control signals. When the simulation starts, the acknowledgement and control signals are not stabilized yet, such that the router is still initializing and setting up the variables and instances, this causes trouble as the source module sees the acknowledgement variable as "0" and thinks the router is ready. It then transmits the flits before the router is even ready to process them, as a result the first two flits are usually lost. After a few clock cycles the system reaches its stable state and correct flow of flits where proper acknowledgement and control signals are performed, and thus the subsequent flits are correctly transmitted and

received by the sink. Now ideally, what we would have expected from the source and router having synchronized clock signals is coordinated execution such that the source and router have the least amount of delay taken to transmit the flit as they both startup at the same time, reducing time mismatches, delays and resulting in more consistent or quicker results(ideal case). However, this unideal behaviour shows that clock synchronization does not guarantee correct communication without the mechanism itself being aligned in the correct manner and time.

4) In order to accurately record time and evaluate the actual delay and performance, a timestamp variable was added to each flit within the 'packet.h', and this same variable is set and modified by the source module through the program line, "sc_time_stamp()". The source module sets the packet variables accordingly, setting this timestamp to the current time(send time), and later these same flits in the packet reach the sink. The sink computes another variable with the current time the sink received the flit(receive time), and the delay is computed as the difference between the receive and send times. These delays are later accumulated and averaged through multiple flits(every 5 flits is 1 packet, that is from head to tail) to compute the average packet delay, and this is done through multiple packets to identify "average packet delay". Do note that the average packet delay is calculated not using each and every flit, but only when the tail flit is received, as it represents the completion of the packet itself. This feature enables proper analysis and evaluation of the network latency.

5) From the beginning, the arbiter module used the destination information from every flit to determine routing and mapping, but this is redundant and is not ideal for the wormhole routing principle. This implementation of the arbiter module was modified so that the decisions for the arbiter

unit is only made when a new connection by the header flit is established. This modification has also been provided by the lab, and is done by checking whether the input is already connected or not; essentially if the flit is not connected, that means it is the header, and it is considered. Once the path is established, the body flits simply follow the reserved path without needing to re-evaluate the destination for each body flit. The connection is then ended when receiving the final tail flit, which indicates that the packet has been sent. This demonstrates the ideal wormhole behaviour and really improved its efficiency.

Methodology:

The main methodology design used within this project is to create a modular and scalable design approach, this was the priority and basis of the methodology behind the construct. Now instead of redesigning the NoC from scratch, the existing 1x2 architecture was extended to make it a 4x4 mesh by replicating routers and constructing an entirely new interconnection scheme. Each router maintains almost the same internal structure with the FIFO buffers, arbiter units, and crossbars, ensuring consistency between designs.

SystemC applications and programs were used to display and determine timing, communication and evaluate the concurrency and functionality of the different modules. It must be understood that each component operates using a clock-drive process, this ensures that data transferred across the mesh topology network is synchronized based on the clock positions the modules get triggered with. The implementation also relies on the methodology of what's called the Wormhole flow control, which preserves packets and ensures efficiency by dividing the packets up into flits, and allowing the routing decision to only be made and determined by the header

flit, so that the rest of the body can follow along with the header flit; this design is effective and very clean as a single flit leads the way ensuring a totally synchronous packet.

Now the implementation of the 4x4 mesh required minimal modifications to the fundamental block modules. The main changes were only restricted to the "main.cpp"(the main cpp file where all the instances, routing, and interconnections occur) for network scaling & replicating instances, and the "source.cpp" was also used and modified to ensure the communication and pathway determining is also functional within this 4x4 mesh. The arbiter logic was also refined previously in the interim phase making sure the wormhole behavior is consistent, eliminating the need for dependence on the body flits at all.

Design:

The 4x4 mesh includes 16 routers arranged in a mesh grid topology. Each router is known to have four directions for its neighbors(North, South, East, and West) and one local IP core with its own dedicated router FIFO buffer input(source/sink and local source FIFO). The design is a uniform structure, allowing it to be scalable and easily extendable.

Main.cpp - Network construct and framework:

The main.cpp file is responsible for instantiating and connecting all components. It includes the 16 sources, sinks, and router modules which are dynamically configured and instantiated using a single loop of 16 iterations. Each router is assigned a unique ID corresponding to its position on the grid.

The key framework of the design for the 4x4 mesh is the connection logic:

- East-West Connections: It needs to be

understood that the routers that are not on the right edge ($col < 3$), have valid east sides with connections to other routers, as such its east output(out3) is connected to adjacent routers(the one beside it to the right) west input(in1), ensuring that the flow goes from a west input port to an east output port. Similarly, routers not on the left edge have left side connections to other routers, as such, it will receive input from their west neighbour through their east input port.

- **North-South Connections:** If a router is not on the bottom edge ($row < 3$), its south output(out4) connects to the north input(in2) of the router below. Similarly for the routers that are not on the top rows ($row > 0$) they receive input from their northern neighbour through their south input port(to the routers north output port)

This systematic strategy makes sure bidirectional communication across the mesh works. It must also be noted that each connection also has a data signal and acknowledgement signals for clean flow control.

To index through the mesh grid topology, ($row = i/4$, $col = i \% 4$) is an easy strategy and simple algorithm that allows systematic connection of the router, without actually having to use addresses with the direct mesh links(connectors between different NoC routers). This design improvement makes sure the implementation is scalable and efficient.

Arbiter.cpp - Routing and Resource Management

The arbiter unit is the most critical component of the router. It determines exactly how the flits are forwarded and distributes directions and route paths for multiple incoming packets that compete for the same output route. The arbiter operates in

the following manner:

1) Input Request Detection:

The arbiter checks within each input port for each respective FIFO buffer to determine if it is empty or not, if its not empty then a flit could be ready:

```
“ if (!req0.read()[4]) ”
```

This bit within the buffer request indicates if a flit is ready and waiting for routing or not.

2) Routing Decisions(XY Dynamic Routing):

The arbiter simply compares the router's ID(corresponds to the location on grid) with the destination address in the flit:

```
“ if(id_col < dst_col) v_req[0]=3;
// go east
else if(id_col > dst_col) v_req[0]=5;
// go west
else if(id_row < dst_row) v_req[0]=4;
// go south
else if(id_row > dst_row) v_req[0]=2;
// go north
else v_req[0]=1; //
local destination “
```

This implements XY routing, since it computes the next direction that it must take to proceed to get closer to the destination. All the values are relative to the grid coordinates(the more east/right the values are the larger compared to west/left, and similarly the south/bottom has higher values in the column than the top/north values in the column)

3) Output Availability Check: The arbiter also uses a bitmask(v_free) to effectively track available output ports. If the desired output is busy, the request is deferred for the moment.

4) Grant Allocation:

If the output is available, a grant is issued:

```
“ grant0.write(1); // set grant ”
```

This very connection is then forwarded to the crossbar for execution and establishment.

5) Wormhole Connection Management:

Once the connection is established through the head flit, it is then maintained using:

```
“ v_connected_input[0]=1; // input 0
is connected
```

```
v_reserved_output[v_req[0]]=1; //
output is reserved “
```

This ensures that the following flits after the head(body flits and tail flit) within the packet follow the same path.

6) Tail Flit Handling:

Once the tail flit is detected, it ends the flow control and clears up the variables used to maintain the body with the head flit:

```
“ if(req0.read()[5]){
v_connected_input[0]=0;v_reserved_
output[v_req[0]]=0;}
// if it is tail flit, reset connection and
reservation} ”
```

This releases the connection, freeing the output port for other input ports to connect to.

This implementation ensures that routing is highly efficient with minimal latency and proper handling of traffic congestion and contention between the data flow streams.

Experimental Results:

The simulation outputs are primarily constructed through the “cout” statements in the source, sink and main modules.

Observed behaviour:

The mapped coordinate for the mesh grid layout can be seen from this output:

Coordinate Mapping Legend:			
ID 0:(0,0)	ID 1:(0,1)	ID 2:(0,2)	ID 3:(0,3)
ID 4:(1,0)	ID 5:(1,1)	ID 6:(1,2)	ID 7:(1,3)
ID 8:(2,0)	ID 9:(2,1)	ID 10:(2,2)	ID 11:(2,3)
ID 12:(3,0)	ID 13:(3,1)	ID 14:(3,2)	ID 15:(3,3)

Figure 3: 4x4 grid map coordinates

1) Packet Transmission:

The source modules generates synthetic traffic, it generates the flits and prints a statement:

New Pkt: 1001 is sent by source: 0 to Destination: 1
New Pkt: 1002 is sent by source: 1 to Destination: 2
New Pkt: 1003 is sent by source: 2 to Destination: 3
New Pkt: 1004 is sent by source: 3 to Destination: 4
New Pkt: 1005 is sent by source: 4 to Destination: 5
New Pkt: 1006 is sent by source: 5 to Destination: 6
New Pkt: 1007 is sent by source: 6 to Destination: 7
New Pkt: 1008 is sent by source: 7 to Destination: 8
New Pkt: 1009 is sent by source: 8 to Destination: 9
New Pkt: 1010 is sent by source: 9 to Destination: 10
New Pkt: 1011 is sent by source: 10 to Destination: 11
New Pkt: 1012 is sent by source: 11 to Destination: 12
New Pkt: 1013 is sent by source: 12 to Destination: 13
New Pkt: 1014 is sent by source: 13 to Destination: 14
New Pkt: 1015 is sent by source: 14 to Destination: 15
New Pkt: 1016 is sent by source: 15 to Destination: 0

Figure 4: Packet Transmission

The statement specifies the packet value/data, the source and oriented destination. As it could be seen the “neighbouring pattern” is implemented, where packets are sent to the adjacent routers. This results in faster and shorter routing paths determined by the routers, which reflect results and values into the sink much quicker.

2) Arbiter’s routing information and action:

The arbiter then prints out the routing scheme determined, it proceeds to give the router address, the destination address, and the selected port it leave through as output within the router(based on N, S, E, W directions)

```
[ROUTER 0 @ (0,0)] Routing Packet to Dest (0,1) -> Selected Port: 3
[ROUTER 1 @ (0,1)] Routing Packet to Dest (0,2) -> Selected Port: 3
[ROUTER 2 @ (0,2)] Routing Packet to Dest (0,3) -> Selected Port: 3
[ROUTER 3 @ (0,3)] Routing Packet to Dest (1,0) -> Selected Port: 5
[ROUTER 4 @ (1,0)] Routing Packet to Dest (1,1) -> Selected Port: 3
[ROUTER 5 @ (1,1)] Routing Packet to Dest (1,2) -> Selected Port: 3
[ROUTER 6 @ (1,2)] Routing Packet to Dest (1,3) -> Selected Port: 3
[ROUTER 7 @ (1,3)] Routing Packet to Dest (2,0) -> Selected Port: 5
[ROUTER 8 @ (2,0)] Routing Packet to Dest (2,1) -> Selected Port: 3
[ROUTER 9 @ (2,1)] Routing Packet to Dest (2,2) -> Selected Port: 3
[ROUTER 10 @ (2,2)] Routing Packet to Dest (2,3) -> Selected Port: 3
[ROUTER 11 @ (2,3)] Routing Packet to Dest (3,0) -> Selected Port: 5
[ROUTER 12 @ (3,0)] Routing Packet to Dest (3,1) -> Selected Port: 3
[ROUTER 13 @ (3,1)] Routing Packet to Dest (3,2) -> Selected Port: 3
[ROUTER 14 @ (3,2)] Routing Packet to Dest (3,3) -> Selected Port: 3
[ROUTER 15 @ (3,3)] Routing Packet to Dest (0,0) -> Selected Port: 5
```

Figure 5: Arbiter Routing Information

The routing mechanism matches perfectly as expected, the selected ports show that the packets are being routed to the right direction by the arbiter that brings the packets closer to the destination intended.

3) Packet Reception:

The sink module then logs the received packets:

```
>>> [SINK 1] RECEIVED Packet from Source 0 | Total Latency: 10000ns
>>> [SINK 2] RECEIVED Packet from Source 1 | Total Latency: 10000ns
>>> [SINK 3] RECEIVED Packet from Source 2 | Total Latency: 10000ns
>>> [SINK 5] RECEIVED Packet from Source 4 | Total Latency: 10000ns
>>> [SINK 6] RECEIVED Packet from Source 5 | Total Latency: 10000ns
>>> [SINK 7] RECEIVED Packet from Source 6 | Total Latency: 10000ns
>>> [SINK 9] RECEIVED Packet from Source 8 | Total Latency: 10000ns
>>> [SINK 10] RECEIVED Packet from Source 9 | Total Latency: 10000ns
>>> [SINK 11] RECEIVED Packet from Source 10 | Total Latency: 10000ns
>>> [SINK 13] RECEIVED Packet from Source 12 | Total Latency: 10000ns
>>> [SINK 14] RECEIVED Packet from Source 13 | Total Latency: 10000ns
>>> [SINK 15] RECEIVED Packet from Source 14 | Total Latency: 10000ns
>>> [SINK 4] RECEIVED Packet from Source 3 | Total Latency: 25000ns
>>> [SINK 8] RECEIVED Packet from Source 7 | Total Latency: 25000ns
>>> [SINK 12] RECEIVED Packet from Source 11 | Total Latency: 25000ns
>>> [SINK 0] RECEIVED Packet from Source 15 | Total Latency: 35000ns
```

Figure 6: Packet Reception from Sink

This clearly confirms that the correct routing mechanism is in place, since all the packets that have been sent through the source have reached its destination sink. The sink also records the time that it reached the sink, while the flit itself stores the time that it was sent, it compares the “sent” and “received” time to compute the total latency.

4) Average Packet Delay:

At the end of the packet being sent, we can see the computed average packet delay, this average delay is the accumulated delays of all 5 flits in the packet, averaged per packet. This delay confronts the latency from a packet wise perspective:

```
===== Average Delay for the packets(5 flits) so far: 50000=====
>>> [SINK 1] RECEIVED Packet from Source 0 | Total Latency: 10000ns
===== Average Delay for the packets(5 flits) so far: 50000=====
>>> [SINK 2] RECEIVED Packet from Source 1 | Total Latency: 10000ns
===== Average Delay for the packets(5 flits) so far: 50000=====
>>> [SINK 3] RECEIVED Packet from Source 2 | Total Latency: 10000ns
===== Average Delay for the packets(5 flits) so far: 50000=====
>>> [SINK 5] RECEIVED Packet from Source 4 | Total Latency: 10000ns
===== Average Delay for the packets(5 flits) so far: 50000=====
>>> [SINK 6] RECEIVED Packet from Source 5 | Total Latency: 10000ns
===== Average Delay for the packets(5 flits) so far: 50000=====
>>> [SINK 7] RECEIVED Packet from Source 6 | Total Latency: 10000ns
===== Average Delay for the packets(5 flits) so far: 50000=====
```

Figure 7: Average Calculated Packet Delay

5) Full-Overview:

The output also shows the corresponding total packets sent and received, as a system overview:

```
End of switch operation...
Total number of packets sent:
source0: 10
source1: 10
source2: 10
source3: 10
source4: 10
source5: 10
source6: 10
source7: 10
source8: 10
source9: 10
source10: 10
source11: 10
source12: 10
source13: 10
source14: 10
source15: 10
```

Figure 8: Total Packets sent by each Source

```
Total number of packets received:
sink0: 10
sink1: 10
sink2: 10
sink3: 10
sink4: 10
sink5: 10
sink6: 10
sink7: 10
sink8: 10
sink9: 10
sink10: 10
sink11: 10
sink12: 10
sink13: 10
sink14: 10
sink15: 10
```

Figure 9: Total Packets Received by each Sink

As it could be seen that those sent by source modules were received by the sink modules, this clearly proves that the results are entirely correct.

[Note: Do note that during the demo, this code was initially not working, and the results did not align at all,

thus the demo showed the incomplete work. This fix was done later and the code executed just fine after this]

The NoC successfully routes the packets across multiple hops using XY routing, and accurately reaches its intended destinations. As well as the arbiter correctly determines appropriate routing paths and actively resolves contentions, managing the packet distribution and input requests, and thus successfully completing its task. The arbiter and the router module also utilizes the wormhole routine ensuring lower latency where a packet is led solely by its head flits, not needing to recompute for every body and tail flit. Do note that sometimes, flit loss may occur due to lack of reset synchronization, but after a few cycles the system stabilizes, that is if clock cycles misalign. The results clearly prove that the 4x4 mesh is operational, functional and efficient under varying traffic patterns and conditions, under heavy contentions as well.

Conclusions:

This project successfully demonstrates the design and simulation of a modular, efficient and scalable Network-on-Chip design. The project really provides a practical understanding of the NoC design, especially how to build it and implement it within a SystemC environment. By extending it from a 1x2 topology to a 4x4 mesh topology, the system clearly showcases its reusability and efficiency of its components modules. This project also focuses on fundamentals like packet-based communication, routing mechanisms, performance evaluations all of which are very important within a modern SoC application. Much of the key modifications like the clock synchronizations, delay records, and refining the wormhole routing implementation, certainly increases the accuracy of the simulation.

Some of the main challenges faced was correctly wiring the interconnections for the mesh grid and reflecting this change within the arbitration mechanism. The arbiter plays a crucial role in making sure the route is calculated and determined properly, effectively avoiding conflicts, while the main_NoC file enables the scalable network and constructs it through the systematic implementation of a grid mesh structure.

Overall, the project provides a strong foundation in NoC design principles including routing, flow control, and network scalability. The results confirm that the 4x4 mesh system functions correctly under different traffic conditions and communication constraints, making it a very successful implementation of a mesh-based 4x4 NoC design.

References:

[1]“sub_objective1,” *Torontomu.ca*, 2025.
<https://www.ee.torontomu.ca/~courses/coe838/lab-project.html>

Appendix:

Main_noc.cpp - FINAL

// main.cpp (4x4 Mesh NoC)

```
#include "systemc.h"
#include <iostream>
#include <stdlib.h>
#include <stdio.h>
#include "packet.h"
#include "source.h"
#include "sink.h"
#include "router.h"
```

```
int sc_main(int argc, char *argv[])
{
```

```
    // -----SIGNALS
```

```
    sc_signal<packet> si_source[16];
    sc_signal<packet> si_sink[16];
```

```
    // Each directional link is a SEPARATE
    signal so each has exactly one driver.
```

```
    // Naming convention: link index =
    row*3+col for E-W (3 per row, 4 rows =
    12)
```

```
    // link index = row*4+col
    for N-S (4 per col-gap, 3 gaps = 12)
```

```
    sc_signal<packet> si_ew[12]; //
    East-bound      : router[i].out3 ->
    router[i+1].in1
```

```
    sc_signal<packet> si_we[12]; //
    West-bound      : router[i+1].out1 ->
    router[i].in3
```

```
    sc_signal<packet> si_ns[12]; //
    South-bound     : router[i].out4 ->
    router[i+4].in2
```

```
    sc_signal<packet> si_sn[12]; //
    North-bound     : router[i+4].out2 ->
    router[i].in4
```

```
    sc_signal<bool> si_ack_src[16],
    si_ack_sink[16];
```

```
    sc_signal<bool> si_ack_ew[12],
    si_ack_we[12];
```

```
    sc_signal<bool> si_ack_ns[12],
    si_ack_sn[12];
```

```
    // dummy signals for unused boundary
    ports
```

```
    sc_signal<packet> si_zero[200];
    sc_signal<bool> si_ack_zero[200];
```

```
    sc_signal<sc_uint<4>> id[16];
```

```
    sc_clock s_clock("S_CLOCK", 125,
    SC_NS, 0.5, 0.0, SC_NS); // source clock
```

```
    sc_clock r_clock("R_CLOCK", 5,
    SC_NS, 0.5, 10.0, SC_NS); // router clock
```

```
    sc_clock d_clock("D_CLOCK", 5,
    SC_NS, 0.5, 10.0, SC_NS); // destination
    clock
```

```
    // Module instantiations follow
```

```
    // Note that modules can be connected by
    hooking up ports
```

```
    // to signals by name or by using a
    positional notation
```

```
    source* src[16];
```

```
    sink* snk[16];
```

```
    router* r[16];
```

```
    // -----INSTANTIATION
```

```
    for(int i = 0; i < 16; i++)
```

```
    {
```

```
        src[i] = new
        source(sc_gen_unique_name("source"));
```

```
        src[i]->packet_out(si_source[i]);
```

```
        src[i]->source_id(id[i]);
```

```
        src[i]->ack_in(si_ack_src[i]);
```

```
        src[i]->CLK(s_clock);
```

```
        snk[i] = new
        sink(sc_gen_unique_name("sink"));
```

```
        snk[i]->packet_in(si_sink[i]);
```

```
        snk[i]->ack_out(si_ack_sink[i]);
```

```
        snk[i]->sink_id(id[i]);
```

```
        snk[i]->selk(d_clock);
```

```
        r[i] = new
```

```

router(sc_gen_unique_name("router"));
r[i]->router_id(id[i]);
r[i]->rclk(r_clock);

// local
r[i]->in0(si_source[i]);
r[i]->out0(si_sink[i]);
r[i]->outack0(si_ack_src[i]);
r[i]->inack0(si_ack_sink[i]);
}

// -----CONNECTION
// hooking up signals to ports by name
// Port mapping: in1/out1=West |
in2/out2=North | in3/out3=East |
in4/out4=South
int z = 0;
int za = 0;

for(int i = 0; i < 16; i++)
{
    int row = i / 4;
    int col = i % 4;

    // ----- WEST input (receives from
the eastbound signal of left neighbour)
-----
    if(col > 0)
    {
        int link = row * 3 + (col - 1); //
E-W link index between col-1 and col
        r[i]->in1(si_ew[link]);
        r[i]->inack1(si_ack_ew[link]);
    }
    else
    {
        r[i]->in1(si_zero[z]);
        r[i]->inack1(si_ack_zero[za]);
        z++; za++;
    }

    // ----- NORTH input (receives
from the southbound signal of upper
neighbour) -----
    if(row > 0)

```

```

{
    int link = (row - 1) * 4 + col; //
N-S link index between row-1 and row
    r[i]->in2(si_ns[link]);
    r[i]->inack2(si_ack_ns[link]);
}
else
{
    r[i]->in2(si_zero[z]);
    r[i]->inack2(si_ack_zero[za]);
    z++; za++;
}

// ----- EAST input (receives from
the westbound signal of right neighbour)
-----
    if(col < 3)
    {
        int link = row * 3 + col; //
W-E link index between col and col+1
        r[i]->in3(si_we[link]);
        r[i]->inack3(si_ack_we[link]);
    }
    else
    {
        r[i]->in3(si_zero[z]);
        r[i]->inack3(si_ack_zero[za]);
        z++; za++;
    }

    // ----- SOUTH input (receives
from the northbound signal of lower
neighbour) -----
    if(row < 3)
    {
        int link = row * 4 + col; // S-N
link index between row and row+1
        r[i]->in4(si_sn[link]);
        r[i]->inack4(si_ack_sn[link]);
    }
    else
    {
        r[i]->in4(si_zero[z]);
        r[i]->inack4(si_ack_zero[za]);
        z++; za++;
    }
}

```

```

    }

    // ----- OUTPUTS

    // ----- WEST output (drives the
    westbound signal toward left neighbour)
    -----
    if(col > 0)
    {
        int link = row * 3 + (col - 1);
        r[i]->out1(si_we[link]); // this
        router is the sole driver of si_we[link]
        r[i]->outack1(si_ack_we[link]);
    }
    else
    {
        r[i]->out1(si_zero[z]);
        r[i]->outack1(si_ack_zero[za]);
        z++; za++;
    }

    // ----- NORTH output (drives the
    northbound signal toward upper neighbour)
    -----
    if(row > 0)
    {
        int link = (row - 1) * 4 + col;
        r[i]->out2(si_sn[link]); // this
        router is the sole driver of si_sn[link]
        r[i]->outack2(si_ack_sn[link]);
    }
    else
    {
        r[i]->out2(si_zero[z]);
        r[i]->outack2(si_ack_zero[za]);
        z++; za++;
    }

    // ----- EAST output (drives the
    eastbound signal toward right neighbour)
    -----
    if(col < 3)
    {
        int link = row * 3 + col;
        r[i]->out3(si_ew[link]); // this

```

```

        router is the sole driver of si_ew[link]
        r[i]->outack3(si_ack_ew[link]);
    }
    else
    {
        r[i]->out3(si_zero[z]);
        r[i]->outack3(si_ack_zero[za]);
        z++; za++;
    }

    // ----- SOUTH output (drives the
    southbound signal toward lower neighbour)
    -----
    if(row < 3)
    {
        int link = row * 4 + col;
        r[i]->out4(si_ns[link]); // this
        router is the sole driver of si_ns[link]
        r[i]->outack4(si_ack_ns[link]);
    }
    else
    {
        r[i]->out4(si_zero[z]);
        r[i]->outack4(si_ack_zero[za]);
        z++; za++;
    }
}

//sc_start(0, SC_NS);
// tracing:
// trace file creation
sc_trace_file *tf =
sc_create_vcd_trace_file("graph");
// External Signals
sc_trace(tf, s_clock, "s_clock");
sc_trace(tf, d_clock, "d_clock");
sc_trace(tf, si_source[0],
"si_source[0]");
sc_trace(tf, si_source[1],
"si_source[1]");
sc_trace(tf, si_source[2],
"si_source[2]");
sc_trace(tf, si_source[3],
"si_source[3]");
sc_trace(tf, si_source[4],

```

```

"si_source[4]");
    sc_trace(tf, si_source[5],
"si_source[5]");
    sc_trace(tf, si_source[6],
"si_source[6]");
    sc_trace(tf, si_source[7],
"si_source[7]");
    sc_trace(tf, si_source[8],
"si_source[8]");
    sc_trace(tf, si_source[9],
"si_source[9]");
    sc_trace(tf, si_source[10],
"si_source[10]");
    sc_trace(tf, si_source[11],
"si_source[11]");
    sc_trace(tf, si_source[12],
"si_source[12]");
    sc_trace(tf, si_source[13],
"si_source[13]");
    sc_trace(tf, si_source[14],
"si_source[14]");
    sc_trace(tf, si_source[15],
"si_source[15]");
    sc_trace(tf, si_sink[0], "si_sink[0]");
    sc_trace(tf, si_sink[1], "si_sink[1]");
    sc_trace(tf, si_sink[2], "si_sink[2]");
    sc_trace(tf, si_sink[3], "si_sink[3]");
    sc_trace(tf, si_sink[4], "si_sink[4]");
    sc_trace(tf, si_sink[5], "si_sink[5]");
    sc_trace(tf, si_sink[6], "si_sink[6]");
    sc_trace(tf, si_sink[7], "si_sink[7]");
    sc_trace(tf, si_sink[8], "si_sink[8]");
    sc_trace(tf, si_sink[9], "si_sink[9]");
    sc_trace(tf, si_sink[10], "si_sink[10]");
    sc_trace(tf, si_sink[11], "si_sink[11]");
    sc_trace(tf, si_sink[12], "si_sink[12]");
    sc_trace(tf, si_sink[13], "si_sink[13]");
    sc_trace(tf, si_sink[14], "si_sink[14]");
    sc_trace(tf, si_sink[15], "si_sink[15]");

for(int i = 0; i < 16; i++)
    id[i].write(i);

cout << endl;
cout <<

```

```

"-----"
"-----" << endl;
    cout << endl << " 4X4 mesh NOC
simulator containing 16 5x5 Wormhole
routers " << endl;

    cout <<
"-----"
"-----" << endl;
    cout << "This is the simulation of a 4x4
Wormhole router. " << endl;
    cout << "We assume the router has 5
input/output ports, with 4 buffers per input
port " << endl;
    cout << "and each flit has 21 bits width "
<< endl;
    cout << " Press \"Return\" key to start
the simulation..." << endl << endl;

    cout << "\nCoordinate Mapping
Legend:" << endl;
    for(int i = 0; i < 16; i++) {
        cout << "ID " << i << ":("
<< i/4 << ", " << i%4 << ")t";
        if((i+1)%4 == 0) cout <<
endl;
    }
    cout << "\n--- STARTING
TRAFFIC FLOW ---\n" << endl;

    getchar();
    sc_start(10*125+124, SC_NS); // during
[(10*125)+124] ns 10 packets will be sent
and received

    sc_close_vcd_trace_file(tf);

    cout << endl << endl <<
"-----"
"-----" << endl;
    cout << "End of switch operation..." <<
endl;
    cout << "Total number of packets sent: "
<< endl;
    for(int i = 0; i < 16; i++)
        cout << " source" << i << ": " <<

```

```

src[i]->pkt_snt << endl;
    cout << "Total number of packets
received: " << endl;
    for(int i = 0; i < 16; i++)
        cout << "   sink" << i << ": " <<
src[i]->pkt_rcv << endl;
        cout <<
"-----"
"-----" << endl;
    cout << " Press \'Return\' key to end the
simulation..." << endl << endl;
    getchar();
    return 0;
}

```

arbiter.cpp - FINAL

```

//arbiter.cpp
#undef SC_INCLUDE_FX

#include "packet.h"
#include "arbiter.h"

void arbiter :: func()
{
    sc_uint<1> v_connected_input[5];
//set when input is connected to an output
    sc_uint<1> v_reserved_output[6];
//set when output is reserved by a input
(one output more for simple coding)
    sc_uint<3> v_req[5];
    sc_uint<5> v_free; // status of
output in term of being free
    sc_uint<4> v_id;
    sc_uint<5> v_arbit;
    sc_uint<15> v_select;
    for(int
i=0;i<5;i++){v_connected_input[i]=0;v_res
erved_output[i]=0;v_req[i]=0;}
    v_free = 31; // '11111'
    v_arbit = 0;
    v_select = 0;

    // functionality

```

```

while( true )
{
    wait();

    sc_uint<2> id_col, id_row,
dst_col, dst_row;
    v_id = arbiter_id.read();
    id_col = v_id.range(1,0);
    id_row = v_id.range(3,2);

    grant0.write(0);
        // reset grant
    grant1.write(0);
        // reset grant
    grant2.write(0);
        // reset grant
    grant3.write(0);
        // reset grant
    grant4.write(0);
        // reset grant
    if (!free_out0.read()) {v_free
= v_free | 1 ; } // set the bit 0 showing the
output 0 is free
    if (!free_out1.read()) {v_free
= v_free | 2 ; }
    if (!free_out2.read()) {v_free
= v_free | 4 ; }
    if (!free_out3.read()) {v_free
= v_free | 8 ; }
    if (!free_out4.read()) {v_free
= v_free | 16 ;}

    v_id = arbiter_id.read();
    if (!req0.read()[4]) //if FIFO
buffer is not empty
    {

if(!v_connected_input[0]) { // if input is not
connected i.e. it is header
        /*if(v_id[0] <
req0.read()[0]) v_req[0]=3; // go to east
        else {
            if(v_id[0] >

```

```

req0.read()[0])v_req[0]=5; //go to west
        else{

if(v_id[1] < req0.read()[1])v_req[0]=4; //
go to south

        else{

            if(v_id[1] >
req0.read()[1])v_req[0]=2; //go to north

            else v_req[0]=1; // that is the
destination

        }

    }*/
    // AFTER (for 4x4):
    //sc_uint<2>

id_col = v_id.range(1,0);
    //sc_uint<2>
id_row = v_id.range(3,2);
    dst_col =
req0.read().range(1,0); // dest bits from req
signal
    dst_row =
req0.read().range(3,2);

    if(id_col <
dst_col) v_req[0]=3; // go east
    else if(id_col
> dst_col) v_req[0]=5; // go west
    else if(id_row
< dst_row) v_req[0]=4; // go south
    else if(id_row
> dst_row) v_req[0]=2; // go north
    else
v_req[0]=1; // local
destination

    }
    switch (v_req[0]) {
    case 1:
v_arbit=v_free & 1; break;
    case 2:
v_arbit=v_free & 2; break;
    case 3:
v_arbit=v_free & 4; break;
    case 4:

```

```

v_arbit=v_free & 8; break;
        case 5:
v_arbit=v_free & 16; break;
        default: break
    ;
    }

if(!v_connected_input[0]) // if input is not
connected

    {
        if
(v_reserved_output[v_req[0]])v_arbit=0; //
if the requested output was reserved, go to
next input

    }

    if(v_arbit!=0){
        // Only print if it's
the start of a new packet (not already
connected)

if(!v_connected_input[0]) {
        cout <<
"[ROUTER " << (int)v_id << " @ (" <<
(int)id_row << "," << (int)id_col << ") "
        <<
"Routing Packet to Dest (" << (int)dst_row
<< "," << (int)dst_col << ") "
        <<
"-> Selected Port: " << (int)v_req[0] <<
endl;
    }

grant0.write(1);
    // set grant

v_select.range(2,0) = v_req[0];
    v_free =
v_free & (~v_arbit); // inactive the related
output

v_connected_input[0]=1; // input 0 is
connected

```



```

v_reserved_output[v_req[0]]=1; // output is
reserved

if(req0.read()[5]){

v_connected_input[0]=0;v_reserved_output
[v_req[0]]=0;} // if it is tail flit, reset
connection and reservation
        }
    }
    if (!req1.read()[4]) //if buffer
is not empty
    {

if(!v_connected_input[1]) {// if input is not
connected i.e. it is header
        /*if(v_id[0]    <
req1.read()[0]) v_req[1]=3; // go to east
        else {
            if(v_id[0]    >
req1.read()[0])v_req[1]=5; //go to west
            else {

if(v_id[1] < req1.read()[1])v_req[1]=4; //
go to south

                                else {

                                    if(v_id[1]    >
req1.read()[1])v_req[1]=2; //go to north

                                    else v_req[1]=1; // that is the
destination
                                }
                            }
                        }*/
                    //sc_uint<2>
id_col = v_id.range(1,0);
                    //sc_uint<2>
id_row = v_id.range(3,2);
                                dst_col =
req1.read().range(1,0); // dest bits from req
signal
                                dst_row =
req1.read().range(3,2);

```

```

                                if(id_col    <
dst_col) v_req[1]=3;    // go east
                                else if(id_col
> dst_col) v_req[1]=5; // go west
                                else if(id_row
< dst_row) v_req[1]=4; // go south
                                else if(id_row
> dst_row) v_req[1]=2; // go north
                                else
v_req[1]=1;    // local
destination
        }

        switch (v_req[1]) {
            case 1:
v_arbit=v_free & 1; break;
            case 2:
v_arbit=v_free & 2; break;
            case 3:
v_arbit=v_free & 4; break;
            case 4:
v_arbit=v_free & 8; break;
            case 5:
v_arbit=v_free & 16; break;
            default: break
;
        }

if(!v_connected_input[1]) // if input is not
connected
    {
        if
(v_reserved_output[v_req[1]])v_arbit=0; //
if the requested output was reserved, go to
next input
    }
    if(v_arbit!=0){
        // if there is
any free output
        // Only print if it's
the start of a new packet (not already
connected)

if(!v_connected_input[0]) {
                                cout    <<
"ROUTER " << (int)v_id << " @ (" <<

```

```

(int)id_row << "," << (int)id_col << ") " <<
"Routing Packet to Dest (" << (int)dst_row
<< "," << (int)dst_col << ") " <<
"-> Selected Port: " << (int)v_req[1] <<
endl;
    }

grant1.write(1);    // set grant

v_select.range(5,3) = v_req[1];
    v_free =
v_free & (~v_arbit); // inactive the related
outputs

v_connected_input[1]=1; // input 1 is
connected

v_reserved_output[v_req[1]]=1; // output is
reserved

if(req1.read()[5]){v_connected_input[1]=0;
v_reserved_output[v_req[1]]=0;} // if it is
tail flit, reset connection and reservation
    }
    if (!req2.read()[4]) //if buffer
is not empty
    {

if(!v_connected_input[2]) {// if input is not
connected i.e. it is header
    /*if(v_id[0] <
req2.read()[0]) v_req[2]=3; // go to east
    else {
        if(v_id[0] >
req2.read()[0])v_req[2]=5; //go to west
        else {

if(v_id[1] < req2.read()[1])v_req[2]=4; //
go to south
            else {

if(v_id[1] >

```

```

req2.read()[1])v_req[2]=2; //go to north

        else v_req[2]=1; // that is the
destination
    }
    }*/
    //sc_uint<2>
id_col = v_id.range(1,0);
    //sc_uint<2>
id_row = v_id.range(3,2);
    dst_col =
req2.read().range(1,0); // dest bits from req
signal
    dst_row =
req2.read().range(3,2);

    if(id_col <
dst_col) v_req[2]=3; // go east
    else if(id_col
> dst_col) v_req[2]=5; // go west
    else if(id_row
< dst_row) v_req[2]=4; // go south
    else if(id_row
> dst_row) v_req[2]=2; // go north
    else
v_req[2]=1; // local
destination
    }

    switch (v_req[2]) {
        case 1:
v_arbit=v_free & 1; break;
        case 2:
v_arbit=v_free & 2; break;
        case 3:
v_arbit=v_free & 4; break;
        case 4:
v_arbit=v_free & 8; break;
        case 5:
v_arbit=v_free & 16; break;
        default: break
    };
    }

if(!v_connected_input[2]) // if input is not

```

```

connected
{
    if
(v_reserved_output[v_req[2]])v_arbit=0; //
if the requested output was reserved, go to
next input
}
if(v_arbit!=0){
    // Only print if it's
the start of a new packet (not already
connected)

if(!v_connected_input[0]) {
    cout <<
"[ROUTER " << (int)v_id << " @ (" <<
(int)id_row << "," << (int)id_col << ") "
    <<
"Routing Packet to Dest (" << (int)dst_row
<< "," << (int)dst_col << ") "
    <<
"-> Selected Port: " << (int)v_req[2] <<
endl;
}

grant2.write(1); // set grant

v_select.range(8,6) = v_req[2];
v_free =
v_free & (~v_arbit); // inactive the related
outputs

v_connected_input[2]=1; // input 1 is
connected

v_reserved_output[v_req[2]]=1; // output is
reserved

if(req2.read()[5]){v_connected_input[2]=0;
v_reserved_output[v_req[2]]=0;} // if it is
tail flit, reset connection and reservation
}
}

if (!req3.read()[4]) //if buffer
is not empty
{

```

```

if(!v_connected_input[3]) { // if input is not
connected i.e. it is header
    /*if(v_id[0] <
req3.read()[0]) v_req[3]=3; // go to east
    else {
        if(v_id[0] >
req3.read()[0])v_req[3]=5; //go to west
        else {

if(v_id[1] < req3.read()[1])v_req[3]=4; //
go to south
        else {

            if(v_id[1] >
req3.read()[1])v_req[3]=2; //go to north

            else v_req[3]=1; // that is the
destination
        }
    }
}*/

//sc_uint<2>
id_col = v_id.range(1,0);
//sc_uint<2>
id_row = v_id.range(3,2);
dst_col =
req3.read().range(1,0); // dest bits from req
signal
dst_row =
req3.read().range(3,2);

if(id_col <
dst_col) v_req[3]=3; // go east
    else if(id_col
> dst_col) v_req[3]=5; // go west
    else if(id_row
< dst_row) v_req[3]=4; // go south
    else if(id_row
> dst_row) v_req[3]=2; // go north
    else
v_req[3]=1; // local
destination
}

switch (v_req[3]) {

```

```

                                case 1:
v_arbit=v_free & 1; break;
                                case 2:
v_arbit=v_free & 2; break;
                                case 3:
v_arbit=v_free & 4; break;
                                case 4:
v_arbit=v_free & 8; break;
                                case 5:
v_arbit=v_free & 16; break;
                                default: break
;
                                }

if(!v_connected_input[3]) // if input is not
connected
{
    if
(v_reserved_output[v_req[3]])v_arbit=0; //
if the requested output was reserved, go to
next input
}
    if(v_arbit!=0){
// Only print if it's
the start of a new packet (not already
connected)

if(!v_connected_input[0]) {
    cout <<
"[ROUTER " << (int)v_id << " @ (" <<
(int)id_row << "," << (int)id_col << ") "
<<
"Routing Packet to Dest (" << (int)dst_row
<< "," << (int)dst_col << ") "
<<
"-> Selected Port: " << (int)v_req[3] <<
endl;
    }

grant3.write(1); // set grant

v_select.range(11,9) = v_req[3];
                                v_free =
v_free & (~v_arbit); // inactive the related
outputs

```

```

v_connected_input[3]=1; // input 3 is
connected

v_reserved_output[v_req[3]]=1; // output is
reserved

if(req3.read()[5]){v_connected_input[3]=0;
v_reserved_output[v_req[3]]=0;} // if it is
tail flit, reset connection and reservation
    }
}

if (!req4.read()[4]) //if buffer
is not empty
{

if(!v_connected_input[4]) {// if input is not
connected i.e. it is header
    /*if(v_id[0] <
req4.read()[0]) v_req[4]=3; // go to east
    else {
        if(v_id[0] >
req4.read()[0])v_req[4]=5; //go to west
        else {

if(v_id[1] < req4.read()[1])v_req[4]=4; //
go to south
        else {

            if(v_id[1] >
req4.read()[1])v_req[4]=2; //go to north

            else v_req[4]=1; // that is the
destination
        }
    }
}*/

//sc_uint<2>
id_col = v_id.range(1,0);
//sc_uint<2>
id_row = v_id.range(3,2);
                                dst_col =
req4.read().range(1,0); // dest bits from req
signal

```

```

                                dst_row    =
req4.read().range(3,2);

                                if(id_col    <
dst_col) v_req[4]=3;    // go east
                                else if(id_col
> dst_col) v_req[4]=5; // go west
                                else if(id_row
< dst_row) v_req[4]=4; // go south
                                else if(id_row
> dst_row) v_req[4]=2; // go north
                                else
v_req[4]=1;    // local
destination
                                }
                                switch (v_req[4]) {
                                    case 1:
v_arbit=v_free & 1; break;
                                    case 2:
v_arbit=v_free & 2; break;
                                    case 3:
v_arbit=v_free & 4; break;
                                    case 4:
v_arbit=v_free & 8; break;
                                    case 5:
v_arbit=v_free & 16; break;
                                    default: break
                                };
                                }

if(!v_connected_input[4]) // if input is not
connected
                                {
                                    if
(v_reserved_output[v_req[4]])v_arbit=0; //
if the requested output was reserved, go to
next input
                                }
                                if(v_arbit!=0){
// Only print if it's
the start of a new packet (not already
connected)
                                if(!v_connected_input[0]) {
cout    <<

```

```

"[ROUTER " << (int)v_id << " @ (" <<
(int)id_row << "," << (int)id_col << ") "
                                <<
"Routing Packet to Dest (" << (int)dst_row
<< "," << (int)dst_col << ") "
                                <<
"-> Selected Port: " << (int)v_req[4] <<
endl;
                                }

grant4.write(1);    // set grant

v_select.range(14,12) = v_req[4];
                                v_free    =
v_free & (~v_arbit); // inactive the related
outputs

v_connected_input[4]=1; // input 4 is
connected

v_reserved_output[v_req[4]]=1; // output is
reserved

if(req4.read()[5]){v_connected_input[4]=0;
v_reserved_output[v_req[4]]=0;} // if it is
tail flit, reset connection and reservation
                                }
                                }
                                aselect.write(v_select);
                                }
}

```

crossbar.cpp - FINAL

```

// crossbar.cpp
#include "packet.h"
#include "crossbar.h"

void crossbar :: func()
{
    packet v_cross0;
    packet v_cross1;
    packet v_cross2;
    packet v_cross3;
    packet v_cross4;

```

```

sc_uint<15> v_config;

// functionality
while( true )
{
    wait();
    v_config = config.read();
    if (i0.event())
    {
        v_cross0 = i0.read();
        switch
        (v_config(2,0)) {
            //
            MODIFIED - AFTER(corrected):
            case 1:
o0.write(v_cross0); break; // local : sink
            case 2:
o2.write(v_cross0); break; // north : out2
            case 3:
o3.write(v_cross0); break; // east : out3
            case 4:
o4.write(v_cross0); break; // south : out4
            case 5:
o1.write(v_cross0); break; // west : out1
            default: cout
<< "-----wrong
destination " <<endl ;break ;
        }
    }
    if (i1.event())
    {
        v_cross1 = i1.read();
        switch
        (v_config(5,3)) {
            //
            MODIFIED - AFTER(corrected):
            case 1:
o0.write(v_cross1); break; // local : sink
            case 2:
o2.write(v_cross1); break; // north : out2
            case 3:
o3.write(v_cross1); break; // east : out3
            case 4:
o4.write(v_cross1); break; // south : out4
            case 5:

```

```

o1.write(v_cross1); break; // west : out1
            default: cout
<< "-----wrong
destination " <<endl; break ;
        }
    }
    if (i2.event())
    {
        v_cross2 = i2.read();
        switch
        (v_config(8,6)) {
            //
            MODIFIED - AFTER(corrected):
            case 1:
o0.write(v_cross2); break; // local : sink
            case 2:
o2.write(v_cross2); break; // north : out2
            case 3:
o3.write(v_cross2); break; // east : out3
            case 4:
o4.write(v_cross2); break; // south : out4
            case 5:
o1.write(v_cross2); break; // west : out1
            default: cout
<< "-----wrong
destination " <<endl; break ;
        }
    }
    if (i3.event())
    {
        v_cross3 = i3.read();
        switch
        (v_config(11,9)) {
            //
            MODIFIED - AFTER(corrected):
            case 1:
o0.write(v_cross3); break; // local : sink
            case 2:
o2.write(v_cross3); break; // north : out2
            case 3:
o3.write(v_cross3); break; // east : out3
            case 4:
o4.write(v_cross3); break; // south : out4
            case 5:
o1.write(v_cross3); break; // west : out1

```

```

                                default: cout
<< "-----wrong
destination " <<endl; break ;
                                }
                                }
                                if (i4.event())
                                {
                                    v_cross4 = i4.read();
                                    switch
(v_config(14,12)) {
                                //
MODIFIED - AFTER(corrected):
                                case 1:
o0.write(v_cross4); break; // local : sink
                                case 2:
o2.write(v_cross4); break; // north : out2
                                case 3:
o3.write(v_cross4); break; // east : out3
                                case 4:
o4.write(v_cross4); break; // south : out4
                                case 5:
o1.write(v_cross4); break; // west : out1
                                default: cout
<< "-----wrong
destination" <<endl ;break ;
                                }
                                }
                                }
}

```

sink.cpp - FINAL

```

// sink.cpp
#include "sink.h"
void sink::receive_data(){
    packet v_packet;
    if ( sclk.event() )
ack_out.write(false);
    if (packet_in.event() )
    {
        pkt_rcv++ ;
        ack_out.write(true);
        v_packet= packet_in.read();

        double   rcv_time   =

```

```

sc_time_stamp().to_double());
        double delay = rcv_time -
v_packet.send_time;

        total_delay += delay;

        if(v_packet.h_t) { // only
count packet at tail
            total_packets++;
            cout << "=====
Average Delay for the packets(5 flits) so
far: " << total_delay / total_packets <<
"===== " << endl;
            cout << "
-----" << endl;
        }
        //cout << "
        New Pkt: " <<
(int)v_packet.data<< " is received from
source: " << (int)v_packet.id << " by sink:
" << (int)sink_id.read() << " With Delay: "
<< (double)delay << endl;
        cout << " >>> [SINK " <<
sink_id.read() << "]" RECEIVED Packet
from Source " << v_packet.id << " | Total
Latency: " << (double)delay << "ns" <<
endl;
    }
}

```

source.cpp - FINAL

```

// source.cpp
#include "source.h"
void source:: func()
{
    packet v_packet_out;
    v_packet_out.data=1000; // e.g.
    v_packet_out.pkt_clk = '0'; // an
imaginary clock for packets
    while(true)
    {
        wait();
        if(!ach_in.read())
        {

```

```

        v_packet_out.data =
v_packet_out.data +source_id.read()+ 1 ; //
made a desired data

        v_packet_out.id    =
source_id.read();

        v_packet_out.dest=
(source_id.read() + 1) % 16;

                                //      assign
destination

                                //if(v_packet_out.id
== 1) goto exclode; //      prevent      from
reciving flits by itself

v_packet_out.pkt_clk=
~v_packet_out.pkt_clk ;      //      add      an
imaginary clock to each flit

v_packet_out.h_t=false;
        pkt_snt++;

v_packet_out.send_time          =
sc_time_stamp().to_double();//Record
TIMESTAMP      of      source      sending      the
packets

if((pkt_snt%5)==0)v_packet_out.h_t=true;
// make tail flit (the packet size is 5)

packet_out.write(v_packet_out);

        cout << "New Pkt: "
<< v_packet_out.data << " is sent by
source: " << source_id.read() << " to
Destination:      "<< v_packet_out.dest
<<endl;
exclode;;

        }

    }
}

```

Main_noc.cpp - INTERIM

```

// main.cpp
#include "systemc.h"
#include <iostream>
#include <stdlib.h>
#include <stdio.h>
#include "packet.h"
#include "source.h"
#include "sink.h"
#include "router.h"

int sc_main(int argc, char *argv[])
{
    sc_signal<packet> si_source[4];
    sc_signal<packet> si_input[4];
    sc_signal<packet> si_zero[16];
    sc_signal<packet> si_sink[4];
    sc_signal<packet> si_output[4];
    sc_signal<bool>
si_ack_src[4],si_ack_ou[4];
    sc_signal<bool>
si_ack_sink[4],si_ack_in[4];
    sc_signal<bool> si_ack_zero[16];
    sc_signal<sc_uint<4>
id0,id1,id2,id3;
    sc_clock      s_clock("S_CLOCK",
125, SC_NS, 0.5, 0.0, SC_NS); // source
clock
    sc_clock      r_clock("R_CLOCK",
125, SC_NS, 0.5, 0.0, SC_NS);      //
router clock
    sc_clock d_clock("D_CLOCK", 5,
SC_NS, 0.5, 10.0, SC_NS); // destination
clock

    // Module instiations follow
    // Note that modules can be

```


connected by hooking up ports
 // to signals by name or by using a
 positional notation

```

    source source0("source0");
    source0(si_source[0],          id0,
si_ack_src[0], s_clock);
    //need codes
    //need codes
    //need codes

```

```

    router router0("router0");
    // hooking up signals to ports by
name

```

```

    router0.in0(si_source[0]);
    router0.in2(si_input[0]);
    router0.in3(si_input[3]);
    router0.in1(si_zero[0]);
    router0.in4(si_zero[1]);

```

```

    router0.router_id(id0);

```

```

    router0.out0(si_sink[0]);
    router0.out2(si_output[0]);
    router0.out3(si_output[3]);
    router0.out1(si_zero[2]);
    router0.out4(si_zero[3]);

```

```

    router0.inack0(si_ack_sink[0]);
    router0.inack2(si_ack_in[0]);
    router0.inack3(si_ack_in[3]);
    router0.inack1(si_ack_zero[0]);
    router0.inack4(si_ack_zero[1]);

```

```

    router0.ouack0(si_ack_src[0]);
    router0.ouack2(si_ack_ou[0]);
    router0.ouack3(si_ack_ou[3]);
    router0.ouack1(si_ack_zero[2]);
    router0.ouack4(si_ack_zero[3]);

```

```

    router0.rclk(r_clock);

```

```

    router router1("router1");
    // hooking up signals to ports by
name

```

```

    router1.in0(si_source[1]);
    router1.in2(si_zero[4]);
    router1.in3(si_input[1]);
    router1.in4(si_output[0]);
    router1.in1(si_zero[5]);

```

```

    router1.router_id(id1);

```

```

    router1.out0(si_sink[1]);
    router1.out2(si_zero[6]);
    router1.out3(si_output[1]);
    router1.out4(si_input[0]);
    router1.out1(si_zero[7]);

```

```

    router1.inack0(si_ack_sink[1]);
    router1.inack2(si_ack_zero[4]);
    router1.inack3(si_ack_in[1]);
    router1.inack4(si_ack_ou[0]);
    router1.inack1(si_ack_zero[5]);

```

```

    router1.ouack0(si_ack_src[1]);
    router1.ouack2(si_ack_zero[6]);
    router1.ouack3(si_ack_ou[1]);
    router1.ouack4(si_ack_in[0]);
    router1.ouack1(si_ack_zero[7]);

```

```

    router1.rclk(r_clock);
    //need 64 code statement

```

```

    //need codes
    sink sink1("sink1");
    sink1(si_sink[1],    si_ack_sink[1],
id1, d_clock);
    //need codes
    //need codes

```

```

//sc_start(0, SC_NS);
// tracing:
    // trace file creation
    sc_trace_file          *tf          =
sc_create_vcd_trace_file("graph");
    // External Signals
    sc_trace(tf, s_clock, "s_clock");
    sc_trace(tf, d_clock, "d_clock");

```

```

        sc_trace(tf,          si_source[0],
"si_source[0]");
        sc_trace(tf,          si_source[0],
"si_source[1]");
        //need codes
        //need codes
        sc_trace(tf,          si_source[0],
"si_sink[0]");
        sc_trace(tf,          si_source[0],
"si_sink[1]");
        //need codes
        //need codes

        id0.write(0);
        id1.write(1);
        //need codes
        //need codes

        cout << endl;
        cout <<
"-----"
"-----" << endl;
        cout << endl << " 1X2 mesh NOC
simulator containing 2 5x5 Wormhole
router " << endl;
        cout <<
"-----"
"-----" << endl;
        cout << "This is the simulation of a
1x2 Wormhole router. " << endl;
        cout << "We assume the router has
5 input/output ports, with 4 buffers per
input port " << endl;
        cout << "and each flit has 21 bits
width " << endl;
        cout << " Press \"Return\" key to
start the simulation..." << endl << endl;

        getchar();
        sc_start(10*125+124, SC_NS); //
during [(10*125)+124] ns 10 packets will
be sent and received

        sc_close_vcd_trace_file(tf);

```

```

        cout << endl << endl <<
"-----"
"-----" << endl;
        cout << "End of switch operation..."
<< endl;
        cout << "Total number of packets
sent: " << source0.pkt_snt<< endl; //need
codes to be added
        cout << "Total number of packets
received: " << sink1.pkt_rcv<<
endl; //need codes to be added
        cout <<
"-----"
"-----" << endl;
        cout << " Press \"Return\" key to end the
simulation..." << endl << endl;
        getchar();
        return 0;
}

```

Source.cpp - INTERIM

```

// source.cpp
#include "source.h"
void source:: func()
{
    packet v_packet_out;
    v_packet_out.data=1000; // e.g.
    v_packet_out.pkt_clk = '0'; // an
imaginary clock for packets
    while(true)
    {
        wait();
        if(!ach_in.read())
        {
            v_packet_out.data =
v_packet_out.data +source_id.read()+ 1 ; //
made a desired data
            v_packet_out.id =
source_id.read();
            v_packet_out.dest=
1;

            // assign destination

```

```

        if(v_packet_out.id
== 1) goto exclude; // prevent from
receiving flits by itself

```

```

v_packet_out.pkt_clk=
~v_packet_out.pkt_clk ; // add an
imaginary clock to each flit

```

```

v_packet_out.h_t=false;
        pkt_snt++;

```

```

v_packet_out.send_time =
sc_time_stamp().to_double();//Record
TIMESTAMP of source sending the
packets

```

```

if((pkt_snt%5)==0)v_packet_out.h_t=true;
// make tail flit (the packet size is 5)

```

```

packet_out.write(v_packet_out);

```

```

        cout << "New Pkt: "
<< v_packet_out.data << " is sent by
source: " << source_id.read() << " to
Destination: " << v_packet_out.dest
<<endl;
        exclude;;
    }
}

```

Sink.h - INTERIM

```

// sink.h
#include "packet.h"
SC_MODULE(sink) {
    sc_in<packet> packet_in; //
input port
    sc_out<bool> ack_out;
    // output port
    sc_in<sc_uint<4>> sink_id;
    sc_in<bool> sclk;
    double total_delay = 0;
    int total_packets = 0;
    int pkt_rcv;

```

```

void receive_data();

```

```

// Constructor
SC_CTOR(sink) {

```

```

SC_METHOD(receive_data); // Method
Process

```

```

        dont_initialize();
        sensitive << packet_in;
        sensitive << sclk.pos();
        pkt_rcv = 0;
    }
};

```

Sink.cpp - INTERIM

```

// sink.cpp

```

```

#include "sink.h"

```

```

void sink::receive_data(){

```

```

    packet v_packet;
    if (sclk.event() )
ack_out.write(false);
    if (packet_in.event() )
    {
        pkt_rcv++ ;
        ack_out.write(true);
        v_packet= packet_in.read();

```

```

        double rcv_time =
sc_time_stamp().to_double();
        double delay = rcv_time -
v_packet.send_time;

```

```

        total_delay += delay;

```

```

        if(v_packet.h_t) { // only
count packet at tail

```

```

            total_packets++;
            cout << "Average
Delay for the packet so far: " <<
total_delay / total_packets << endl;
        }
        cout << "

```

```

        New      Pkt:      "      <<
(int)v_packet.data<< " is received from
source: " << (int)v_packet.id << " by sink:
" << (int)sink_id.read() << " With Delay: "
<< (double)delay << endl;

    }
}

```

packet.h-INTERIM

```

// packet.h file
#ifndef PACKET
#define PACKET
#include "systemc.h"
struct packet {
    sc_uint<11> data;
    sc_uint<4> id; //
packet source ID
    sc_uint<4> dest; //
packet destination ID
    sc_uint<1> pkt_clk; // bit
for changing the new packet condition
    sc_uint<1> h_t;
// header or tail flit("1" represents
tail)

    int send_time;
    inline bool operator == (const packet&
rhs) const
    {
        return (rhs.data == data &&
rhs.id == id && rhs.dest == dest &&
rhs.pkt_clk == pkt_clk && rhs.h_t == h_t);
    }
};

inline
ostream&
operator << ( ostream& os, const packet& a
)
{
    os << "streaming of struct packet not
implemented";
}

```

```

        return os;
    }

    inline
    void
    #if defined(SC_API_VERSION_STRING)
        sc_trace( sc_trace_file* tf, const packet&
a, const std::string& name )
    #else
        sc_trace( sc_trace_file* tf, const packet&
a, const sc_string& name )
    #endif
    {
        sc_trace( tf, a.data, name + ".data" );
        sc_trace( tf, a.id, name + ".id" );
        sc_trace( tf, a.dest, name + ".dest" );
        sc_trace( tf, a.pkt_clk, name + ".pkt_clk"
);
        sc_trace( tf, a.h_t, name + ".h_t" );
    }
}

```

#endif

arbiter.cpp-INTERIM

```

//arbiter.cpp
#undef SC_INCLUDE_FX

#include "packet.h"
#include "arbiter.h"

void arbiter :: func()
{
    sc_uint<1> v_connected_input[5];
//set when input is connected to an output
    sc_uint<1> v_reserved_output[6];
//set when output is reserved by a input
(one output more for simple coding)
    sc_uint<3> v_req[5];
    sc_uint<5> v_free; // status of
output in term of being free
    sc_uint<4> v_id;
    sc_uint<5> v_arbit;
    sc_uint<15> v_select;
    for(int

```

```

i=0;i<5;i++){v_connected_input[i]=0;v_reserved_output[i]=0;v_req[i]=0;}
    v_free = 31; // '11111'
    v_arbit = 0;
    v_select = 0;

    // functionality

    while( true )
    {
        wait();
        grant0.write(0);
            // reset grant
        grant1.write(0);
            // reset grant
        grant2.write(0);
            // reset grant
        grant3.write(0);
            // reset grant
        grant4.write(0);
            // reset grant
        if (!free_out0.read()) {v_free
= v_free | 1 ; } // set the bit 0 showing the
output 0 is free
        if (!free_out1.read()) {v_free
= v_free | 2 ; }
        if (!free_out2.read()) {v_free
= v_free | 4 ; }
        if (!free_out3.read()) {v_free
= v_free | 8 ; }
        if (!free_out4.read()) {v_free
= v_free | 16 ;}

        v_id = arbiter_id.read();
        if (!req0.read()[4]) //if FIFO
buffer is not empty
        {

if(!v_connected_input[0]) { // if input is not
connected i.e. it is header
            if(v_id[0] <
req0.read()[0]) v_req[0]=3; // go to east
            else {
                if(v_id[0] >

```

```

req0.read()[0])v_req[0]=5; //go to west
                else{

if(v_id[1] < req0.read()[1])v_req[0]=4; //
go to south
                else{

                    if(v_id[1] >
req0.read()[1])v_req[0]=2; //go to north

                    else v_req[0]=1; // that is the
destination
                }
            }
        }
        switch (v_req[0]) {
            case 1:
                v_arbit=v_free & 1; break;
            case 2:
                v_arbit=v_free & 2; break;
            case 3:
                v_arbit=v_free & 4; break;
            case 4:
                v_arbit=v_free & 8; break;
            case 5:
                v_arbit=v_free & 16; break;
            default: break
        }
        ;

        if(!v_connected_input[0]) // if input is not
connected
        {
            if
(v_reserved_output[v_req[0]])v_arbit=0; //
if the requested output was reserved, go to
next input
        }

        if(v_arbit!=0){

grant0.write(1);
            // set grant

```

```

v_select.range(2,0) = v_req[0];
v_free =
v_free & (~v_arbit); // inactive the related
output

v_connected_input[0]=1; // input 0 is
connected

v_reserved_output[v_req[0]]=1; // output is
reserved

if(req0.read()[5]){

v_connected_input[0]=0;v_reserved_output
[v_req[0]]=0;} // if it is tail flit, reset
connection and reservation
    }
    }
    if (!req1.read()[4]) //if buffer
is not empty
    {

if(!v_connected_input[1]) { // if input is not
connected i.e. it is header
    if(v_id[0] <
req1.read()[0]) v_req[1]=3; // go to east
    else {
        if(v_id[0] >
req1.read()[0])v_req[1]=5; //go to west
        else {

if(v_id[1] < req1.read()[1])v_req[1]=4; //
go to south

else {

        if(v_id[1] >
req1.read()[1])v_req[1]=2; //go to north

        else v_req[1]=1; // that is the
destination
    }
    }
    }
}
switch (v_req[1]) {

```

```

case 1:
v_arbit=v_free & 1; break;
case 2:
v_arbit=v_free & 2; break;
case 3:
v_arbit=v_free & 4; break;
case 4:
v_arbit=v_free & 8; break;
case 5:
v_arbit=v_free & 16; break;
default: break
;
}

if(!v_connected_input[1]) // if input is not
connected
{
    if
(v_reserved_output[v_req[1]])v_arbit=0; //
if the requested output was reserved, go to
next input
}
if(v_arbit!=0){
    // if there is
any free output
grant1.write(1); // set grant

v_select.range(5,3) = v_req[1];
v_free =
v_free & (~v_arbit); // inactive the related
outputs

v_connected_input[1]=1; // input 1 is
connected

v_reserved_output[v_req[1]]=1; // output is
reserved

if(req1.read()[5]){v_connected_input[1]=0;
v_reserved_output[v_req[1]]=0;} // if it is
tail flit, reset connection and reservation
}
}
if (!req2.read()[4]) //if buffer

```

```

is not empty
{

if(!v_connected_input[2]) { // if input is not
connected i.e. it is header
    if(v_id[0] <
req2.read()[0]) v_req[2]=3; // go to east
    else {
        if(v_id[0] >
req2.read()[0])v_req[2]=5; //go to west
        else {

if(v_id[1] < req2.read()[1])v_req[2]=4; //
go to south
                                else {

                                if(v_id[1] >
req2.read()[1])v_req[2]=2; //go to north

                                else v_req[2]=1; // that is the
destination
                                }
                                }
                                }

                                switch (v_req[2]) {
                                case 1:
v_arbit=v_free & 1; break;
                                case 2:
v_arbit=v_free & 2; break;
                                case 3:
v_arbit=v_free & 4; break;
                                case 4:
v_arbit=v_free & 8; break;
                                case 5:
v_arbit=v_free & 16; break;
                                default: break
                                ;
                                }

if(!v_connected_input[2]) // if input is not
connected
{
    if

```

```

(v_reserved_output[v_req[2]])v_arbit=0; //
if the requested output was reserved, go to
next input
                                }
                                if(v_arbit!=0){

grant2.write(1); // set grant

v_select.range(8,6) = v_req[2];

v_free =
v_free & (~v_arbit); // inactive the related
outputs

v_connected_input[2]=1; // input 1 is
connected

v_reserved_output[v_req[2]]=1; // output is
reserved

if(req2.read()[5]){v_connected_input[2]=0;
v_reserved_output[v_req[2]]=0;} // if it is
tail flit, reset connection and reservation
                                }
                                }

                                if (!req3.read()[4]) //if buffer
is not empty
                                {

if(!v_connected_input[3]) { // if input is not
connected i.e. it is header
    if(v_id[0] <
req3.read()[0]) v_req[3]=3; // go to east
    else {
        if(v_id[0] >
req3.read()[0])v_req[3]=5; //go to west
        else {

if(v_id[1] < req3.read()[1])v_req[3]=4; //
go to south
                                else {

                                if(v_id[1] >
req3.read()[1])v_req[3]=2; //go to north

```

```

        else v_req[3]=1; // that is the
destination
    }
}

}

switch (v_req[3]) {
    case 1:
v_arbit=v_free & 1; break;
    case 2:
v_arbit=v_free & 2; break;
    case 3:
v_arbit=v_free & 4; break;
    case 4:
v_arbit=v_free & 8; break;
    case 5:
v_arbit=v_free & 16; break;
    default: break
;
}

if(!v_connected_input[3]) // if input is not
connected
{
    if
(v_reserved_output[v_req[3]])v_arbit=0; //
if the requested output was reserved, go to
next input
}
if(v_arbit!=0){

grant3.write(1); // set grant

v_select.range(11,9) = v_req[3];
v_free =
v_free & (~v_arbit); // inactive the related
outputs

v_connected_input[3]=1; // input 3 is
connected

v_reserved_output[v_req[3]]=1; // output is
reserved

```

```

if(req3.read()[5]){v_connected_input[3]=0;
v_reserved_output[v_req[3]]=0;} // if it is
tail flit, reset connection and reservation
}

}

if (!req4.read()[4]) //if buffer
is not empty
{
    if(!v_connected_input[4]) {// if input is not
connected i.e. it is header
        if(v_id[0] <
req4.read()[0]) v_req[4]=3; // go to east
        else {
            if(v_id[0] >
req4.read()[0])v_req[4]=5; //go to west
            else {

if(v_id[1] < req4.read()[1])v_req[4]=4; //
go to south

                else {

                    if(v_id[1] >
req4.read()[1])v_req[4]=2; //go to north

                        else v_req[4]=1; // that is the
destination
                }
            }
        }
    }

}

switch (v_req[4]) {
    case 1:
v_arbit=v_free & 1; break;
    case 2:
v_arbit=v_free & 2; break;
    case 3:
v_arbit=v_free & 4; break;
    case 4:
v_arbit=v_free & 8; break;
    case 5:
v_arbit=v_free & 16; break;
    default: break

```



```

;
    }

if(!v_connected_input[4]) // if input is not
connected
    {
        if
(v_reserved_output[v_req[4]])v_arbit=0; //
if the requested output was reserved, go to
next input
    }
    if(v_arbit!=0){

grant4.write(1);    // set grant

v_select.range(14,12) = v_req[4];
                    v_free      =
v_free & (~v_arbit); // inactive the related
outputs

v_connected_input[4]=1; // input 4 is
connected

v_reserved_output[v_req[4]]=1; // output is
reserved

if(req4.read()[5]){v_connected_input[4]=0;
v_reserved_output[v_req[4]]=0;} // if it is
tail flit, reset connection and reservation
    }
    }
    aselect.write(v_select);
    }
}

```